North Ridge Highway Intersection Upgrade

IFC model register

Model file: north-ridge-highway.ifc. Schema: IFC4. 7 named objects. Length units are metres. Local vertical datum is 670.000 m; add this to model Z coordinates to obtain the project elevations.

This register is extracted from the IFC object descriptions and property sets. It is a searchable companion to the conceptual model. Project drawings and governing standards remain the design evidence. Geometry includes explicitly identified visualization assumptions and is not a certified survey or a construction model.

Object / IFC class	Register page
RD-EB - Eastbound through lane / IfcCivilElement	2
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Model SHA-256: 53ea60239fd97177e362ccc461bc626c92d8730431ddca15b73b446d4dafbf5a

Search by object tag, nominal diameter, cover, clearance, lane width, sight distance or pavement layer. Each object page retains its IFC GlobalId and the source drawing or report used for its properties.

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IFC object RD-EB - Eastbound through lane

IFC object RD-EB is the eastbound through lane. Its width is 3.5 m on the 12.0 m paved surface. The design speed of IFC object RD-EB is 100 km/h. Available eastbound stopping sight distance is 165 m. The profile crest is 671.875 m at chainage 0+300. Final geometry and grade-corrected SSD require review.

IFC class	IfcCivilElement
GlobalId	0svb8hfuPHSuo0Bt8GKC4e
Width	3.5 m
Design Speed	100 km/h
Paved Surface Width	12.0 m
Available Eastbound SSD	165 m
Initial SSDscreen	185 m, level-road basis; grade effects outstanding
Profile	L250 m; grades +3% / -3%; K41.67 m/%
Geometry	Station 0+100 to 0+500; surface display thickness 2 mm is illustrative

Source trace

nrh_geometric-design-plan-profile_ifr_2026-09-12.pdf · PDF page 3. Model status: Issued for review.

nrh_geometric-design-plan-profile_ifr_2026-09-12.pdf · PDF page 1 (related design basis).

Concept geometry; verify against cited design documents.

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IFC object RD-WB - Westbound through lane

IFC object RD-WB is the westbound through lane. Its width is 3.5 m on the 12.0 m paved surface. The design speed of IFC object RD-WB is 100 km/h. Available eastbound stopping sight distance is 165 m. The profile crest is 671.875 m at chainage 0+300. Final geometry and grade-corrected SSD require review.

IFC class	IfcCivilElement
GlobalId	3Bvd0L2cbGGw_qBTAV7Ctw
Width	3.5 m
Design Speed	100 km/h
Paved Surface Width	12.0 m
Available Eastbound SSD	165 m
Initial SSDscreen	185 m, level-road basis; grade effects outstanding
Profile	L250 m; grades +3% / -3%; K41.67 m/%
Geometry	Station 0+100 to 0+500; surface display thickness 2 mm is illustrative

Source trace

nrh_geometric-design-plan-profile_ifr_2026-09-12.pdf · PDF page 1. Model status: Issued for review.

Concept geometry; verify against cited design documents.

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IFC object SH-S - South paved shoulder

IFC object SH-S is the south paved shoulder. Its width is 2.5 m on the 12.0 m paved surface. The design speed of IFC object SH-S is 100 km/h. Available eastbound stopping sight distance is 165 m. The profile crest is 671.875 m at chainage 0+300. Final geometry and grade-corrected SSD require review.

IFC class	IfcCivilElement
GlobalId	3i$5uabxXVgRDfjhtSuqOA
Width	2.5 m
Design Speed	100 km/h
Paved Surface Width	12.0 m
Available Eastbound SSD	165 m
Initial SSDscreen	185 m, level-road basis; grade effects outstanding
Profile	L250 m; grades +3% / -3%; K41.67 m/%
Geometry	Station 0+100 to 0+500; surface display thickness 2 mm is illustrative

Source trace

nrh_geometric-design-plan-profile_ifr_2026-09-12.pdf · PDF page 1. Model status: Issued for review.

Concept geometry; verify against cited design documents.

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IFC object SH-N - North paved shoulder

IFC object SH-N is the north paved shoulder. Its width is 2.5 m on the 12.0 m paved surface. The design speed of IFC object SH-N is 100 km/h. Available eastbound stopping sight distance is 165 m. The profile crest is 671.875 m at chainage 0+300. Final geometry and grade-corrected SSD require review.

IFC class	IfcCivilElement
GlobalId	2DVEcjv2XJRv7Cz_G21rfu
Width	2.5 m
Design Speed	100 km/h
Paved Surface Width	12.0 m
Available Eastbound SSD	165 m
Initial SSDscreen	185 m, level-road basis; grade effects outstanding
Profile	L250 m; grades +3% / -3%; K41.67 m/%
Geometry	Station 0+100 to 0+500; surface display thickness 2 mm is illustrative

Source trace

nrh_geometric-design-plan-profile_ifr_2026-09-12.pdf · PDF page 1. Model status: Issued for review.

Concept geometry; verify against cited design documents.

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IFC object PAV-ACP - Trial asphalt layer

IFC object PAV-ACP is the trial asphalt layer. The thickness of IFC object PAV-ACP is 110 mm. Pavement draft inputs are 7,200 AADT, 10% trucks and 1.0% annual non-compounded growth. This trial section is not a verified structural design.

IFC class	IfcCivilElement
GlobalId	10A6yafYDTvwHilhGmBh1G
Thickness	110 mm
Status	Trial option; structural adequacy not established
Traffic AADT	7200
Truck Share	10%
Growth	1.0% annual non-compounded
Subgrade CBR	5%

Source trace

nrh_pavement-design-memo_rev0_2026-09-19.pdf · PDF page 1. Model status: Issued for review.

Concept geometry; verify against cited design documents.

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IFC object PAV-BASE - Trial granular base

IFC object PAV-BASE is the trial granular base layer. Its thickness is 250 mm below the trial 110 mm asphalt layer. Loading must be reconciled with the traffic assessment; no ESAL-based adequacy is claimed.

IFC class	IfcCivilElement
GlobalId	227ISfgILK$gmlXEfPV7QE
Thickness	250 mm
Status	Trial option; structural adequacy not established
Traffic AADT	7200
Truck Share	10%
Growth	1.0% annual non-compounded

Source trace

nrh_pavement-design-memo_rev0_2026-09-19.pdf · PDF page 1. Model status: Issued for review.

Concept geometry; verify against cited design documents.

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IFC object ACC-N - Industrial access context

IFC object ACC-N is the illustrative north industrial access. Traffic assessment inputs are 7,800 AADT, 14% heavy vehicles and 2.0% annual non-compounded growth. The 2047 twenty-year combined traffic horizon is missing. Access geometry is illustrative and has not been designed.

IFC class	IfcCivilElement
GlobalId	3F1P_DBv5TAfXydDnsyL44
Arrangement	North T-intersection; no south access
Station	0+300
Geometry	Illustrative access envelope; levels, throat width and tie-in unresolved
TIAAADT	7800
TIAHeavy Vehicles	14%
TIAGrowth	2.0% annual non-compounded
TIAMissing Horizon	2047

Source trace

nrh_traffic-impact-assessment_rev1_2026-09-15.pdf · PDF page 1. Model status: Issued for review.

Concept geometry; verify against cited design documents.